Score of a Bounded Distributions

May 3, 2025

1 Bounded Distribution

Let $x_0 \sim B(a,b)$ be a bounded distribution, with p.d.f. $p(x)$. We will derive the score function for p_t, where p_t is the probability density function of x_t, with:

$$x_t = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1-\bar{\alpha}_t}\varepsilon \quad \text{where} \varepsilon \sim \mathcal{N}(0, I)$$

So, p_t has the closed form:

$$p_t(x|y) = \frac{1}{\sqrt{2\pi}\sigma_t} \exp\left(-\frac{(x-\sqrt{\bar{\alpha}_t}y)^2}{2\sigma_t^2}\right)$$

$$\implies \quad p_t(x) = \frac{1}{\sqrt{2\pi}\sigma_t} \int_a^b p(y) \exp\left(-\frac{(x-\sqrt{\bar{\alpha}_t}y)^2}{2\sigma_t^2}\right) dy$$

Then, the score function is given by:

$$\begin{aligned}
s_t(x) &= \nabla_x \log p_t(x) \\
&= \frac{\nabla_x p_t(x)}{p_t(x)} \\
&= \frac{\displaystyle\int_a^b \left(\frac{\sqrt{\bar{\alpha}_t}y - x}{\sigma_t^2}\right) p(y) \exp\left(-\frac{(x-\sqrt{\bar{\alpha}_t}y)^2}{2\sigma_t^2}\right) dy}{\displaystyle\int_a^b p(y) \exp\left(-\frac{(x-\sqrt{\bar{\alpha}_t}y)^2}{2\sigma_t^2}\right) dy} \\
&= \frac{1}{\sigma_t^2}\left(\sqrt{\bar{\alpha}_t} m(x) - x\right) \\
&= \frac{1}{1-\bar{\alpha}_t}\left(\sqrt{\bar{\alpha}_t} m(x) - x\right)
\end{aligned}$$

where $m(x)$ is give by:

$$m(x) = \frac{\displaystyle\int_a^b y p(y) \exp\left(-\frac{(x-\sqrt{\bar{\alpha}_t}y)^2}{2\sigma_t^2}\right) dy}{\displaystyle\int_a^b p(y) \exp\left(-\frac{(x-\sqrt{\bar{\alpha}_t}y)^2}{2\sigma_t^2}\right) dy}$$

Now, we will analyse 3 cases:

- $\bar{\alpha}_t \to 0$
- $x >> b$
- $x << a$

1.1 Case 1: $\bar{\alpha}_t \to 0$

In this case, we simply get:

$$s_t(x) \approx -x$$

1.2 Case 2: $x >> b$

Recall $m(x)$:

$$m(x) = \frac{\displaystyle\int_a^b y p(y) \exp\left(-\frac{(x-\sqrt{\bar{\alpha}_t}y)^2}{2\sigma_t^2}\right) dy}{\displaystyle\int_a^b p(y) \exp\left(-\frac{(x-\sqrt{\bar{\alpha}_t}y)^2}{2\sigma_t^2}\right) dy}$$

Let $q(y|x)$ be the conditional distribution of y given x. Then, we can write:

$$q(y|x) = \frac{p(y) \exp\left(-\frac{(x-\sqrt{\bar{\alpha}_t}y)^2}{2\sigma_t^2}\right)}{\displaystyle\int_a^b p(y) \exp\left(-\frac{(x-\sqrt{\bar{\alpha}_t}y)^2}{2\sigma_t^2}\right) dy}$$

So, we can write $m(x)$ as:

$$m(x) = \int_a^b y q(y|x) dy = \mathbb{E}[y|x]$$

Now if $x >> b$, then $q(y|x)$ has most of its mass at $y = b$. So, we can approximate:

$$m(x) \approx b$$

Thus, we have:

$$s_t(x) = \frac{\sqrt{\bar{\alpha}_t} \cdot b - x}{\sigma_t^2}$$

1.3 Case 3: $x << a$

In this case, we can use the same argument as in Case 2. Thus:

$$s_t(x) = \frac{\sqrt{\bar{\alpha}_t} \cdot a - x}{\sigma_t^2}$$